

Building Energy Prediction: 2-Stage XGBoost Implementation

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1 Objective

Develop a machine learning pipeline to predict building energy consumption (Electricity, District Heating, District Cooling) at 10-minute intervals. A **2-stage architecture** specifically addresses cooling sparsity: Stage 1 detects ON/OFF states via classifier; Stage 2 predicts magnitude via regressor trained only on ON samples.

2 Data & Features

2.1 Integration

Four InfluxDB CSV streams merged: energy consumption (kWh), 4-zone indoor temperatures, DNI (W/m^2), shading control. Chronological preprocessing preserves causality; 70/30 train-test split prevents lookahead bias. Dataset: 1,429 records at 10-min intervals (Jan–Dec 2026), final features: 30.

2.2 Feature Set

- **Environmental** (4): T_{ext} , DNI, $T_{\text{air,mean}}$, $\Delta T = T_{\text{air}} - T_{\text{ext}}$
- **Temporal** (4): $\sin(2\pi h/24)$, $\cos(2\pi h/24)$, $\sin(2\pi d/365)$, $\cos(2\pi d/365)$
- **Interactions** (2): $\text{DNI} \times \Delta T$, $\text{DNI} \times T_{\text{ext}}$
- **Lags** (12): Electricity, Heating, Cooling at $t - 1, 6, 12, 24$ steps
- **Rolling** (4): Cooling mean over 1, 6, 12, 24 steps
- **State** (4): Binary count of cooling ON in 1, 6, 12, 24-step windows

All features causal; no data leakage.

3 Model Architecture

Electricity & Heating: XGBoost regressors ($n = 1200$, $lr = 0.03$, $d = 6$, $\lambda = 1.0$).

Cooling (2-Stage):

1. **Classifier:** XGBoost binary ($n = 700$, $d = 5$, class-weighted for 13.87% ON imbalance)
2. **Regressor:** ON-only samples; log-transform; peak-aware weights $w = \min(3, 1 + 2y/p_{90})$
3. **Hard Gate:** $\hat{y}_{\text{cool}} = \begin{cases} \hat{y}_{\text{mag}} & p_{\text{ON}} \geq \tau \\ 0 & p_{\text{ON}} < \tau \end{cases}$, $\tau \approx 0.70$

4 Results

4.1 Primary Energy

$$E_{\text{primary}} = \frac{E_{\text{heat}}}{0.98} \times 1.0 + \frac{E_{\text{cool}}}{5.4} \times 2.17 + E_{\text{elec}} \times 2.17$$

4.2 Test Metrics

Target	R^2	MSE
Electricity	0.9612	1.50e-06
District Heating	0.8431	7.88e-06
District Cooling	0.8345	2.53e-05
Total Primary Energy	0.9121	2.45e-05

Cooling ON-only (43/422 samples): $R^2 = 0.537$, MSE=2.4e-04 — reflects sparsity challenge.

4.3 Top Features

1. $T_{\text{air,mean}}$ — thermal state
2. Cooling rolling max (6-step) — peak detection
3. Cooling rolling mean (6-step) — trend
4. $\cos(\text{hour})$ — occupancy cycle
5. Cooling ON-recent — system persistence

5 Prediction Quality

- **Electricity:** Tight diagonal alignment ($R^2 = 0.961$), minimal bias.
- **Heating:** Good, slight extremes spread ($R^2 = 0.843$).
- **Cooling:** Global 0.835 (good OFF accuracy); ON-only 0.537 (peak magnitude challenge).
- **Total Energy:** Strong aggregate (0.912) confirms balanced multi-target.

6 Physical Coherence & Deployment

6.1 Feature Justification

- **Environmental:** Core HVAC drivers per thermodynamics
- **Temporal:** Occupancy/solar geometry patterns
- **Lags [1,6,12,24]:** Immediate response, hourly, daily periodicity, building inertia
- **State indicator:** HVAC persistence (doesn't toggle instantly)

6.2 Real-World Viability

- ✓ Chronological split prevents lookahead bias
- ✓ Features from standard building sensors
- ✓ Cooling sparsity handled explicitly
- ! Cold-start: First 24 steps need warm-up or synthetic bootstrap
- ! Building-specific: Retrain if HVAC retrofitted or use pattern changes

7 Conclusions

A 2-stage XGBoost ensemble achieves strong energy prediction: Electricity 96.1%, total primary energy 91.2%. Cooling excels at state detection (0.835) but challenges remain for peak magnitude (0.537), expected for rare events. Feature engineering balances physics-based interpretability with ML performance. All 30 features contribute meaningfully. Model deployable with cold-start awareness and periodic retraining. Trained model saved (joblib) for real-time control or forecasting systems.